
Lab 01: Setting Up Your AWS Account, IAM User, and Access Keys

Course: Cloud Computing **Estimated Time:** 60-90 minutes **Difficulty:** Beginner (no prior AWS experience required)

1. Learning Objectives

By the end of this lab, you will be able to:

1. Create a personal AWS account (Free Tier).
 2. Secure the **root user** of your account with Multi-Factor Authentication (MFA).
 3. Create an **IAM user** for daily lab work (never using the root user for regular tasks).
 4. Enable MFA on the IAM user.
 5. Generate an **access key** for the IAM user to use with the AWS CLI/SDK in later labs.
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2. Why This Matters (Read Before You Start)

- The **root user** is the “owner” of your AWS account. It has unrestricted access to everything, including billing. If it is compromised, an attacker can rack up huge charges or delete all your resources.
- Best practice (and AWS’s own recommendation) is to **lock down the root user with MFA** and then **never use it for daily work**. Instead, you create a limited-privilege **IAM user** for yourself and use that for labs.
- **Access keys** are long-term credentials (like a username/password pair for programmatic access). They must be treated like a password — never shared, never committed to GitHub, never emailed.

⚠ **Cost Warning:** AWS Free Tier covers most services used in this course, but you are still responsible for any charges. Always set a budget alert (Section 6) and terminate/delete resources when a lab is done.

3. Prerequisites

Before starting, make sure you have:

- ☐ A personal email address **not already used** for an AWS account (use your student email or a personal one — do not use a shared/family email).
 - ☐ A valid credit or debit card (required by AWS for identity verification; you will not be charged unless you exceed Free Tier limits).
 - ☐ A phone number that can receive SMS or phone calls (for identity verification).
 - ☐ A **smartphone** with an authenticator app installed. Recommended (pick one):
 - Google Authenticator
 - Microsoft Authenticator
 - Authy
 - ☐ A modern web browser (Chrome, Firefox, or Edge).
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4. Part A — Create Your AWS Account

1. Go to <https://aws.amazon.com/> and click **Create an AWS Account**.
2. Enter your **email address** and choose an **AWS account name** (e.g., `yourname-cloudlab`).
3. Verify your email using the code AWS sends you.
4. Create a **strong root user password** (12+ characters, mix of upper/lower case, numbers, symbols). Save it in a password manager — do not write it in plain text on your laptop.
5. Choose **Personal** as the account type.
6. Fill in your contact information (name, phone number, address).
7. Enter your **payment information**. This is required even for Free Tier accounts.
8. Complete **identity verification**: AWS will send a verification code via SMS or automated call.
9. Select the **Basic Support Plan (Free)**.
10. Wait for the confirmation screen — your account is now created.

✓ **Checkpoint:** You should be able to sign in at <https://console.aws.amazon.com/> using the **root user** email and password.

5. Part B — Secure the Root User with MFA

Do this immediately after creating your account, before anything else.

1. Sign in to the AWS Console as the **root user**.
2. In the top-right search bar, type **IAM** and open the **IAM** service.
3. On the IAM dashboard, look for the **Security recommendations** panel and find **Add MFA for root user** (or navigate to **My Security Credentials** from the account name menu, top-right).
4. Click **Assign MFA device**.
5. Give the device a name, e.g., `root-mfa-device`.
6. Choose **Authenticator app** as the MFA type.
7. Open your authenticator app on your phone and **scan the QR code** shown on screen (or enter the setup key manually).
8. The app will start generating a rotating 6-digit code. Enter **two consecutive codes** from the app into the AWS console as prompted.
9. Click **Add MFA**.

✓ **Checkpoint:** Sign out and sign back in as root. You should now be prompted for an MFA code after entering your password.

📌 **Tip:** If you lose your phone, you may lose access to your account. Some students choose to also register a second MFA device (AWS allows multiple MFA devices for the root user).

6. Part B.1 — Set a Budget Alert (Recommended)

While still signed in as root:

1. Search for **Billing and Cost Management** in the console search bar.
2. Go to **Budgets** → **Create budget**.
3. Choose a **Cost budget**, set a monthly amount (e.g., \$5), and add your email for alerts.
4. Save the budget.

This ensures you get an email warning if you accidentally leave a resource running.

7. Part C — Create an IAM User for Lab Work

You should now stop using the root user for daily tasks. Instead, create an IAM user.

1. While signed in as root, go to the **IAM** service.
2. In the left sidebar, click **Users** → **Create user**.
3. Enter a username, e.g., `lab-user` or `firstname-lastname`.
4. Check **Provide user access to the AWS Management Console** (so you can log in via a browser too).
5. Choose **I want to create an IAM user** (not “federated”).
6. Set a custom console password.
7. Click **Next**.
8. On the **Permissions** screen, choose one of the following:
 - **Attach policies directly**
 - AmazonEC2FullAccess
 - AmazonScalingFullAccess
 - IAMFullAccess
 - AmazonS3FullAccess
 - AmazonDynamoDBFullAccess
9. Click **Next**, review the details, then click **Create user**.
10. On the confirmation screen, **download the .csv file** or copy the console sign-in link, username, and password. You will need this to log in as this user going forward.

✓ **Checkpoint:** Open a new **private/incognito browser window**, go to the IAM sign-in link, and log in as your new IAM user.

8. Part D — Enable MFA on the IAM User

Repeat the MFA setup, but this time for your IAM user (not root):

1. Sign in to the console as your **IAM user** (not root).
2. Click your username in the top-right corner → **Security credentials**.
3. Under **Multi-factor authentication (MFA)**, click **Assign MFA device**.
4. Name the device, e.g., `lab-user-mfa`.
5. Choose **Authenticator app**, scan the QR code with your authenticator app.
6. Enter two consecutive codes to confirm.
7. Click **Add MFA**.

✓ **Checkpoint:** Sign out and sign back in as the IAM user — you should be prompted for both password and MFA code.

9. Part E — Generate an Access Key for the IAM User

Access keys let you (or the AWS CLI/SDK) interact with AWS programmatically — you will need this in future labs.

1. While signed in as the **IAM user**, go to **Security credentials** (same page as above).
 2. Scroll to the **Access keys** section and click **Create access key**.
 3. Select the use case that matches your situation — for this course, choose **Command Line Interface (CLI)**.
 4. Check the confirmation box acknowledging the recommended alternatives, then click **Next**.
 5. (Optional) Add a description tag, e.g., `cloud-computing-lab`.
 6. Click **Create access key**.
 7. You will see:
 - **Access key ID** (e.g., `AKIA...`)
 - **Secret access key** (shown only once!)
 8. Click **Download .csv file** to save both values securely. Do **not** paste these into chat apps, email, or public repositories.
- ✓ **Checkpoint:** You now have an access key pair. You will use these in a future lab to configure the AWS CLI (`aws configure`).

🔒 **Critical Security Rule:** If you ever accidentally expose your access key (e.g., commit it to GitHub), immediately go back to this screen, **deactivate** and **delete** the key, then generate a new one.

10. Submission Checklist

Submit a short lab report (1 page) including:

- ☐ Screenshot of your IAM dashboard showing **“MFA on root user: Enabled”** (blur/hide any sensitive info).
- ☐ Screenshot of your IAM **Users** list showing your lab user.
- ☐ Screenshot showing MFA enabled on the IAM user.
- ☐ Confirmation (text only — **do not include actual key values**) that you successfully generated an access key, e.g., *“Access key created: AKIA***...*** (last 4 chars: XXXX)”*.
- ☐ One paragraph reflecting on why root user MFA and least-privilege IAM users matter for cloud security.

11. Common Issues & Troubleshooting

Issue	Likely Cause	Fix
MFA codes rejected	Phone clock out of sync	Enable automatic time sync in your phone's date/time settings
Can't see “Create access key” button	Signed in as root, not IAM user	Root discourages access keys by default; sign in as the IAM user
Verification call/SMS not received	Wrong country code or carrier delay	Try the alternate verification method (call vs. SMS)
Locked out after losing MFA device	MFA device lost/reset	Contact AWS Support via account recovery; this can take time, so keep backup codes if offered
Access key not visible after creation	Left the page before saving	Delete the key and create a new one — the secret is shown only once

12. Key Terms Glossary

- **Root user** – The original, all-powerful account identity created at sign-up.
- **IAM (Identity and Access Management)** – AWS service for managing users, groups, and permissions.
- **MFA (Multi-Factor Authentication)** – A second verification step (beyond password) using a time-based code.
- **Access key** – A programmatic credential pair (Access Key ID + Secret Access Key) used by CLI/SDK tools instead of a password.
- **Least privilege** – Granting only the permissions needed to do a task, nothing more.